

TACKLING THE COMPLEX CHALLENGES IN THE TREATMENT OF BLOOD CANCER IN NIGERIA



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Introduction

Hematological malignancies also referred to as blood cancers are heterogeneous group of neoplasm of clonal origin, arising from haemopoietic tissue (bone marrow and reticuloendothelial tissue).

Blood cancers are a group of cancers that arise from a malignant transformation of cells of the bone marrow or lymphatic system. These cancers include but not limited to leukemia, lymphomas and myeloma.

Incidence

Blood cancers make up a fifth of the most commonly occurring cancers and the second leading cause of mortality. Leukemia was the fifth most common cause of cancer deaths in men and the sixth most common in women in the United States in a 5-year study between 2007 and 2011. Total cancer cases in Nigeria in 2018 was about 115,950 with cancer deaths recorded to be 70,327. Leukemia accounts for 2.3% of the most common cancers in Nigeria in 2018 and contributed 3.2% mortality while Non-Hodgkin Lymphoma accounts in 2018 and contributed 3.2% of mortality cancer cases in Nigeria in 2020 among age group 0-14 years was about 11,837 cases with acute Lymphoid leukemia being 1,317 and Hodgkin Lymphoma, 622 while Burkitt Lymphoma was 823 cases recorded.

In a sixteen-year retrospective analysis of the prevalence and type of hematological malignancies (blood cancers) among adults in a tertiary hospital in Jos-Nigeria, Egesie et al reported a total of 1706 cases of cancers diagnosed during this period. Of among the adult patient population in their study was 19.3%, with male (56.7%) dominance. Leukemia occurred the most (64.5%) followed by Lymphoma (30%) and multiple myeloma (5.4%). Chronic leukemia occurred more frequently than acute leukemia and non-Hodgkin's lymphoma was more common than Hodgkin's lymphoma. The mean SD were 44 17 years for leukemia and 43 15 years for lymphoma.

Aetiology/ predisposing factors

The aetiology of hematological cancers is not known. However, some factors have been identified to increase the risk of their occurrence. Such factors include infections with viruses such as human immunodeficiency virus (HIV), Epstein Barr Virus, Human T- Lymphotropic virus; chronic bacterial infection like helicobacter pylori; immunosuppression or immunodeficiency states, radiation and chemicals such as benzene. These factors are thought to be involved in a complex interplay of chronic antigenetic stimulation resulting in development of these malignancies.

The incidence of blood cancers has been shown to vary according to age, gender, geographic region, and histologic subtypes, while cure rates can vary according to region and maybe impacted by prompt and accurate diagnosis, treatment availability and access to care.

Burden of blood cancers outcomes

Blood cancers are major burden to afflicted patients and their families medically, financially and psychologically. The cost of managing these disorders is very expensive. Individuals diagnosed of these diseases receive treatment through out-of-pocket expenses and are therefore unable to commence, continue treatment as a result of financial constraint. For those who successfully completed the recommended regimen, family income is depleted. The outcomes of the sufferers of blood cancer can vary with the time of presentation, stage of the disease at presentation, type of the disease, other co-morbidities, availability and affordability of drugs, availability and access to palliative and supportive medical care. In Nigeria there is a wide gap between cancer burden and a comprehensive cancer control radiotherapy and palliative/ supportive care.

Challenges in Management of blood cancers in Nigeria

Late presentation: a report of the retrospective study conducted at Ibadan in Oyo state, Nigeria by Fasola et al; revealed that 42(65%) out of a total of 64 patients with multiple myeloma presented at stage 3 Durie and salmon staging. They also reported financial constraint alongside their late presentation. Factors responsible for late presentation includes: ignorance and passiveness to seek for medical care, concern about cost of care in health facilities, indulging in self-medications, herbal drugs usage, faith healers, delays in referral, non-availability of diagnostic and therapeutic facilities. Patients with blood cancers who present late to treatment facilities have increased mortality. However, early presentation with early detection has been reported to result in better outcome.

Poor economic status: Mmeremikwu et al., 2005 in their retrospective study done in Calabar, Cross River state, Nigeria reported that 21(50%) of the children with Burkitt Lymphoma could not afford the diagnostic confirmatory test. Also, 10(25%) of the studied population could not afford the diagnostic confirmatory test. Also, it was observed in their study that 8(20%) of the patients could not afford chemotherapy treatment while 13(31.7%) of the caregivers discontinued treatment after signing against medical advice. The reason being financial constraints.

Limitation of the number of investigations done to aid diagnosis

In a retrospective study done in Port Harcourt, Rivers state by Kurubo et al., 2019 revealed that only 3.1% of 129 patients with haematological malignancies they studied were able to carry out immune-phenotyping requested for diagnosis. This is due to high cost of the investigation.

Difficulty in diagnosis and classification of lymphomas due to use of only morphology without immunochemistry was reported by Akinde et al., 2016 Lagos/Lagos state in their cross-sectional study. Out of the 116 cases of lymphomas, only 32 (27.6%) were correctly diagnosed and subtyped by morphology alone, 43 (46%) was diagnosed to be lymphomas but could not be subtyped by morphology.

Unaffordability and unavailability and of novel drugs: due to high cost of treatment, none of the patients received stem cell transplantation as part of treatment despite having indications for it. The mean total monthly cost of chemotherapy drug per patient was N69,321,21+84,686.43(USD189.92+232.02). In the population studied about 30% of the patients did not receive any form of chemotherapy due to financial constraints, early mortality or being lost to follow up.

Non-availability of health insurance: out of pocket payment for treatment of the most blood cancers are usually not sustainable. A retrospective study carried out by Asuquo et al, 2014 in Calabar, Cross River State, reported that high cost of managing blood cancers constitutes a major contributing factor to poor compliance and survival. Cost of care ranged from USD192.94 to 846.35 for chemotherapy regimen and USD 12,400.49 to 124,699.41 for targeted regimen. Multiple myeloma treatment with chemotherapy could cost between USD 916, 47 to 3,061.77 and USD 39,870.59 for targeted drugs. Acute leukemia patients chemotherapy treatment cost could be between USD 240.00 to 1,058.20. Chronic leukemia patients chemotherapy treatment could cost between USD 240.00 to 1,058.20 chronic leukemia patients chemotherapy treatment could cost between USD194.12 to 3,411.77 for chemotherapy and USD 12,400.59 to 12,352.94 for targeted regimen. Diagnostic work up and follow up expenditure ranged from USD1, 110.59 to 1,878.90. Patients will require USD 947.41 to 1,801.48 for supportive care.

Studies have shown also reported that low level of awareness of blood cancers among healthcare providers at all levels of healthcare.

Overcoming the challenges of treatment of blood cancers in Nigeria

1. The adoption of health insurance and health planning might to a large extent increase access to standard care.
2. Government should subsidize the cost of diagnosis and management of blood cancers.
3. Provision of diagnostic equipment and ensure implantation of health policy that is favorable to patients.
4. Measures to encourage early referral can reduce complication.
5. Medical education and public awareness.
6. Need for more collaborations with non-governmental organizations for funding, awareness, research and information sharing.

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